

Failure mode measured on the VCS	VCS measured evidence proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.000 position-regime F single record Vanilla saliency & IG faithfulness = 0.000 on the position regime; the whole gradient/correlational bucket = 0.068 (n=9). Provably 0 at a strobe-timed argmax readout (\$3.2 oracle).	Shattered / saturated gradients; non-differentiable argmax & index readouts (analogue refs G1, G2).	proven here ↓ analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record Program-randomization test: saliency cosine = 1.000 to the real map while 95% of RAM changed -> sanity check FAILS. Constant Jacobian on the content path, no ReLU.	Saliency unchanged under model- & label-randomization; behaves like an edge detector (analogue ref S1).	proven here ↓ analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.02, 0.29] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116 (spurious tuning) -- both vs the exact \$3.2 oracle.	Spurious Granger edges; 'present ≠ used' tuning & attention-as-explanation (analogue refs C1, C2).	proven here ↓ analogue there
4 Single-unit ablation has an interaction blind-spot	0.414 connectome F 95% CI [0.08, 0.78] Single-unit lesion importance is faithful (0.987) yet MISSES the wiring: the connectome graph is recovered at only 0.414; SAE-feature ablations move y in 0 of the audited cells.	Single-unit ablation misleading; computation distributed across units (analogue refs D1, D2).	proven here ↓ analogue there
5 SAE / probe: present-not-used + polysemanticity	0.162 selectivity F 95% CI [0.08, 0.24] SAEs match a known variable (100% of candidates in 4/6 games, min 83%) yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are decodable-not-causal -> selectivity F = 0.162.	Control-task selectivity gap; superposition / polysemantic neurons (analogue refs P1, P2).	proven here ↓ analogue there
6 Baseline-sensitivity of Integrated / Expected Gradients	0.034 EG F (all-regime) 95% CI [0.00, 0.10] IG magnitude is baseline-sensitive (corr_sensitivity up to 0.999 across games); a baseline equal to the content byte kills IG entirely; Expected Gradients lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input (analogue refs B1, B2).	proven here ↓ analogue there

VCS evidence: proven in this benchmark (vs the exact \$3.2 intervention oracle)

NN analogue: documented in the literature, not proven by this benchmark

Headline (position regime): causal/intervention 0.412 vs gradient/correlational 0.068 (95% CI 0.390 / 0.070)

0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).

Analogue references (documented in the literature, not measured here):
G1 Balduzzi et al. 2017; G2 Sundararajan et al. 2017 (IG). S1 Adebayo et al. 2018. C1 Anand et al. 2019; C2 Jain & Wallace 2019.
D1 Morcos et al. 2018; D2 Leavitt & Morcos 2020. P1 Hewitt & Liang 2019; P2 Elhage et al. 2022. B1 Kindermans et al. 2019; B2 Sturmfels et al. 2020.